TECHNOLOGY REFERENCE

DormDash

Every language, framework, library, service, and tool used to build, test, secure, and ship this mobile-first web application.

DOCUMENT DATE	REPOSITORY	PRODUCTION	PLATFORMS
31 October 2026	github.com/rselvak/dormdash	dormdash-dusky.vercel.app	Web / PWA (mobile-first)

01 Summary

DormDash is a Next.js application deployed to Vercel through GitHub Actions. The stack is deliberately small: one framework, one styling engine, one authentication provider, and one hosting platform. Everything is written in TypeScript and validated on every pull request before it can reach `main`.

Application
Next.js 16 App Router with React 19 server and client components, compiled by Turbopack.

Interface
Tailwind CSS v4 with a custom light/dark theme, responsive layouts, and a mobile bottom tab bar.

Identity
Clerk handles sign-in, sign-up, sessions, OAuth, and MFA. Protected routes call `auth.protect()`.

Delivery
GitHub Actions runs CI and deploys to Vercel preview (test) and production environments.

02 Application stack

TECHNOLOGY	VERSION	ROLE
Next.js	`16.4.3`	Full-stack React framework. App Router, server components, route handlers, file-based metadata, and the production build pipeline.
React	`19.3.1`	UI library. Server components render on the server; client components provide interactivity.
React DOM	`19.3.1`	DOM renderer paired with React for the browser and server-rendered output.
TypeScript	`5.9.3`	Language for all application code. `strict` mode is enabled; target ES2017, bundler module resolution.

TECHNOLOGY	VERSION	ROLE
Tailwind CSS	`4.3.3`	Utility-first styling. Theme tokens defined with `@theme inline` in `globals.css`; automatic dark mode via `prefers-color-scheme`.
@tailwindcss/turbopack	`4.3.3`	Tailwind integration for Turbopack, registered in `next.config.ts` so CSS compiles inside the Next.js build.
@clerk/nextjs	`7.9.23`	Authentication SDK. Provider, middleware, hosted sign-in/sign-up UI, session handling, and server-side authorisation.
@clerk/backend	`3.26.5`	Transitive Clerk server SDK used by `@clerk/nextjs` to verify sessions and call the Clerk API.
Geist & Geist Mono	`via next/font`	Self-hosted variable fonts loaded by `next/font/google`, exposed as `--font-geist-sans` and `--font-geist-mono`. No external font requests at runtime.
Web App Manifest	`Next.js metadata route`	Generates `/manifest.webmanifest` so the site installs to a phone home screen as a PWA.

Runtime and language settings

- Node.js **26.13.0** locally; CI pins Node **22**; `package.json` requires `>=20.9.0`.
- npm **11.20.0** with a committed `package-lock.json`; CI installs with `npm ci`.
- Path alias `@/*` maps to `./src/*`; generated route types are included from `.next/types`.
- Turbopack is the default bundler for both `next dev` and `next build` in Next.js 16.
- Request interception lives in `src/proxy.ts` — Next.js 16 renamed `middleware.ts` to `proxy.ts`.

03 Authentication and security

TECHNOLOGY	VERSION	ROLE
Clerk	`7.9.23`	Hosted identity provider: email + password, Google OAuth, session cookies, and multi-factor authentication.
Clerk middleware	`clerkMiddleware()`	Runs in `src/proxy.ts`, protects the session context, and injects a per-request Content-Security-Policy nonce.
Route-level authorisation	`auth.protect()`	Each protected page calls `await auth.protect()` on the server. Unauthenticated visitors are redirected to sign-in with a return URL.
Content Security Policy	`nonce-based`	Restricts scripts, styles, fonts, images, and frames to known origins, including the Clerk Frontend API and Turnstile challenge endpoints.
HTTP security headers	`next.config.ts`	`X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: strict-origin-when-cross-origin`, `Permissions-Policy` disabling camera, microphone, geolocation and browsing topics, `Strict-Transport-Security` for two years with preload.
Secret management	–	Clerk keys live in `.env.local` (gitignored), Vercel environment variables, and GitHub Actions secrets. No keys are committed; git history was scanned before the repo was made public.

> **Defence in depth.** Authorisation is enforced per route on the server, not by hiding UI. The middleware handles session context and CSP; the page itself re-checks the session with `auth.protect()`, so a new route is safe by default the moment it opts in.

04 Tooling and code quality

TECHNOLOGY	VERSION	ROLE
ESLint	`9.39.5`	Static analysis for JavaScript and TypeScript, run as `npm run lint` in CI.
eslint-config-next	`16.4.3`	Next.js rule set covering React hooks, accessibility, and framework-specific patterns.
Vitest	`5.0.5`	Unit test runner. Tests live in `tests/**/*.test.ts`, run in a Node environment, and resolve the `@` alias.

TECHNOLOGY	VERSION	ROLE
Vite	`8.3.5`	Transitive engine behind Vitest — handles TypeScript transforms and module resolution for tests.
TypeScript compiler	`5.9.3`	`tsc --noEmit` for type checking. The script runs `next typegen` first because route prop types are generated.
@types/node	`24.19.2`	Node.js type definitions.
@types/react / @types/react-dom	`19.x`	React type definitions matching the React 19 runtime.
npm audit	`11.20.0`	Dependency vulnerability gate in CI at `--audit-level=high`. Six moderate dev-only advisories are known and accepted.
Dependabot	`GitHub native`	Weekly npm and GitHub Actions updates, grouped into minor/patch PRs. Major bumps for TypeScript, ESLint, Next.js and React are held back until the tooling supports them.
CodeQL	`action v4`	GitHub semantic code analysis for JavaScript and TypeScript. Runs on push, on pull requests, and weekly. Skips Dependabot PRs, whose token is read-only.

Quality gates on every pull request



The job is named **“Lint, typecheck, test, build”** and is a required status check on `main`. All six steps run on `ubuntu-latest` with Node 22, so a clean runner reproduces the same result as a developer machine.

05 CI/CD and hosting

TECHNOLOGY	VERSION	ROLE
Git	`2.56.0`	Version control. Branches: `main` (production) and `test` (preview).
GitHub	–	Public source repository, pull requests, Actions, environments, and the branch ruleset.
GitHub Actions	–	CI and deployment orchestration. Four workflows: `ci.yml`, `codeql.yml`, `deploy-test.yml`, `deploy-prod.yml`.
actions/checkout	`v7`	Clones the repository into the runner.
actions/setup-node	`v7`	Installs Node 22 with npm caching.
github/codeql-action	`init + analyze @ v4`	Initialises and runs CodeQL, then uploads results to GitHub code scanning.
GitHub CLI	`2.103.3`	Repository, secrets, environments, ruleset, and workflow management from the terminal.
Vercel	–	Hosting and CDN. Builds the Next.js output and serves Production and Preview environments.
Vercel CLI	`65.9.0`	Used in CI to `pull` environment variables, `build`, and `deploy`. Also used locally to link the project and manage variables.

Deployment flows

WORKFLOW	TRIGGER	WHAT IT DOES
CI	Push to `main` / `test`, all pull requests	Install with `npm ci`, lint, typecheck, unit tests, production build, audit.
Deploy to Test	Push to `test`, or manual	Pull the Vercel preview environment, build, deploy a preview, then HTTP smoke test the URL.
Deploy to Production	**Manual only** (`workflow_dispatch`)	Pull the production environment, build, deploy with `--prod`, then smoke test.
CodeQL	Push, pull request, weekly schedule	Static security analysis and upload of results to code scanning.

> **Production is a deliberate human decision.** Merging to `main` runs CI but does not ship. A release is started from *Actions → Deploy to Production → Run workflow*. This works on every GitHub plan, including the free tier where required reviewers are unavailable on private repositories.

Branch protection and environments

- Ruleset `protect-main` on `main`: no deletion, no force-push, required status check, and the branch must be up to date before merging.
- GitHub environments `test` and `production` record deployment history and expose a deployment URL.
- Deploy jobs self-skip with a notice if `VERCEL_TOKEN`, `VERCEL_ORG_ID`, or `VERCEL_PROJECT_ID` are missing, so CI never fails for lack of hosting credentials.
- Vercel Deployment Protection is disabled so the pipeline's smoke tests and the public site are reachable; Clerk still guards every private route.

06 Environments

ENVIRONMENT	URL	NOTES
Local	`http://localhost:3000`	`npm run dev` with Turbopack; keys from `.env.local`.
Production	`dormdash-dusky.vercel.app`	Vercel production target, deployed only by the manual workflow.
Test	`Vercel preview URL`	A unique preview deployment per push to `test`; the URL is printed in each Actions run.
Pull request	`Vercel preview URL`	Per-PR previews when the Vercel GitHub integration is connected.

Environment variables

VARIABLE	PURPOSE
`NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY`	Public Clerk key, inlined into the browser bundle.
`CLERK_SECRET_KEY`	Server-side Clerk key. Never committed and never exposed to the client.
`NEXT_PUBLIC_CLERK_SIGN_IN_URL`	Path of the sign-in page.
`NEXT_PUBLIC_CLERK_SIGN_UP_URL`	Path of the sign-up page.
`NEXT_PUBLIC_CLERK_SIGN_IN_FALLBACK_REDIRECT_URL`	Where to land after sign-in when no return URL is given.
`NEXT_PUBLIC_CLERK_SIGN_UP_FALLBACK_REDIRECT_URL`	Where to land after sign-up.
`VERCEL_ORG_ID / VERCEL_PROJECT_ID`	GitHub secrets identifying the Vercel project.
`VERCEL_TOKEN`	GitHub secret authorising Vercel CLI deploys.

07 Mobile and interface

TECHNOLOGY	VERSION	ROLE
CSS viewport units & safe areas	`native CSS`	`env(safe-area-inset-*)` padding helpers keep content clear of iPhone notches and Android gesture bars.
Next.js viewport export	`Viewport API`	Sets `width=device-width`, `initial-scale=1`, and `viewport-fit=cover`.
Responsive layout	`Tailwind breakpoints`	Single-column mobile layout that expands to a multi-column desktop layout; a bottom tab bar provides thumb-reachable navigation on small screens.
PWA manifest	`metadata route`	App name, theme colour, display mode, and icons, enabling “Add to Home Screen” on iOS and Android.
Accessibility	`eslint-config-next`	ARIA labels, keyboard focus styles, and semantic landmarks, with a CI lint gate on accessibility rules.

08 Local development toolchain

TOOL	VERSION	ROLE
Node.js	`26.13.0`	Local JavaScript runtime. CI uses the Node 22 LTS line.
npm	`11.20.0`	Package manager and script runner; lockfile committed for reproducible installs.
Git	`2.56.0`	Version control and branch workflow.
GitHub CLI	`2.103.3`	Authenticates as `rselvak`; manages the repo, secrets, and workflows.
Vercel CLI	`65.9.0`	Project linking, environment variables, and manual deploys. Invoked through `npx` so no global install is required.

NPM scripts

SCRIPT	COMMAND
`npm run dev`	`next dev`
`npm run build`	`next build`
`npm run start`	`next start`
`npm run lint`	`eslint`
`npm run typecheck`	`next typegen && tsc --noEmit`
`npm test`	`vitest run`
`npm run test:watch`	`vitest`

09 Current status and known gaps

Working end to end

- ✓ Login and protected routes
- ✓ Responsive mobile UI and PWA manifest
- ✓ CI on every pull request
- ✓ Test and production deploys with smoke tests
- ✓ Manual release gate and branch protection

Not yet in place

Database — PostgreSQL and Prisma are planned but not wired up.
Clerk production keys — the live site still runs development keys.
PNG icons — an SVG icon exists; iOS needs 192/512 PNG and an apple-touch-icon.
Input validation — Zod is intended for server-side schema validation.

Upgrade policy. Next.js, React, TypeScript, and ESLint majors are treated as planned migrations rather than automatic pull requests. Dependabot is configured to raise only minor and patch updates for them, because a major bump can break `typescript-eslint` or `eslint-config-next` before those tools catch up.